

Title: Retail Sales Data Processing and
Business Insights Generation

Company: ABC Retail Solutions

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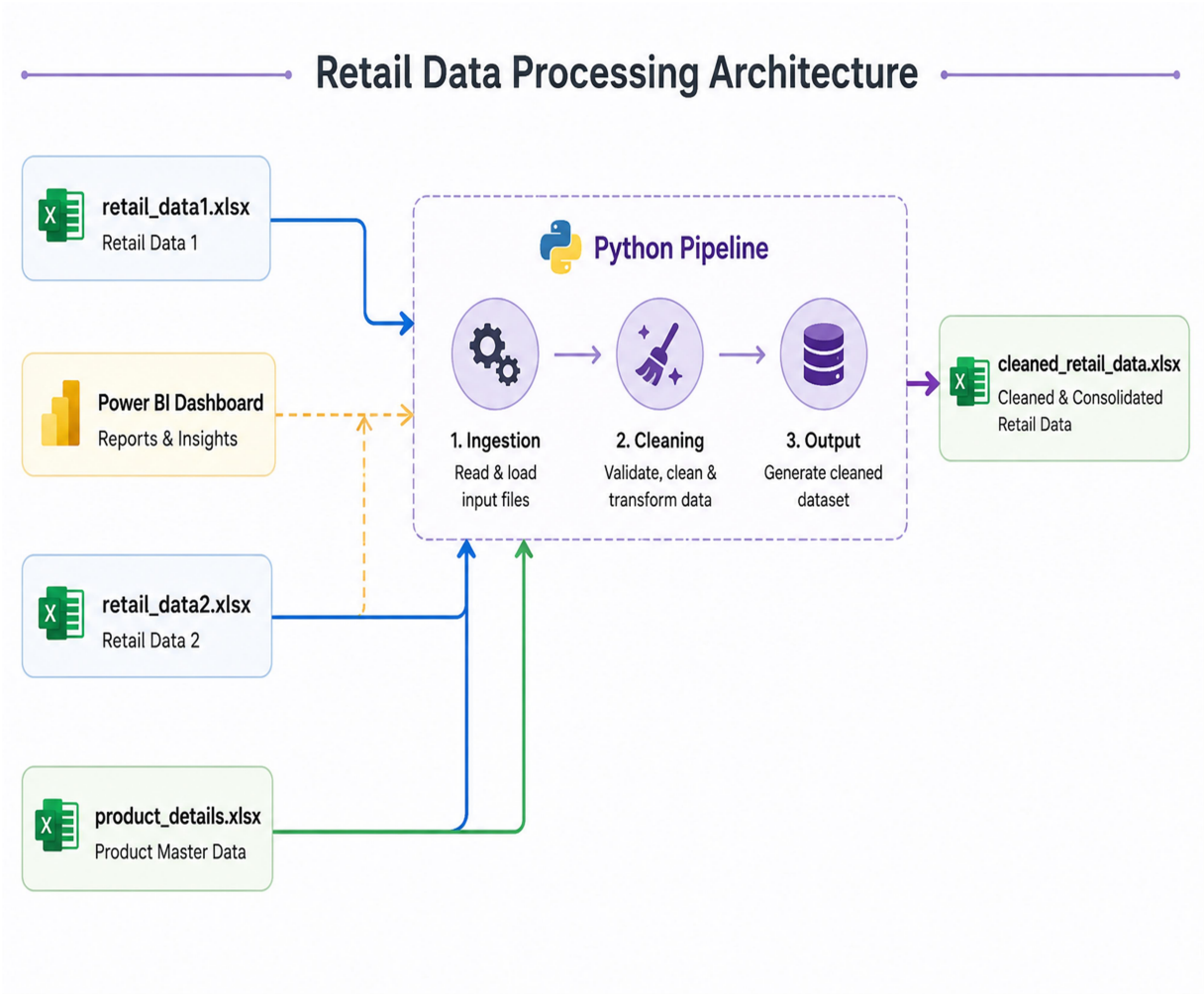
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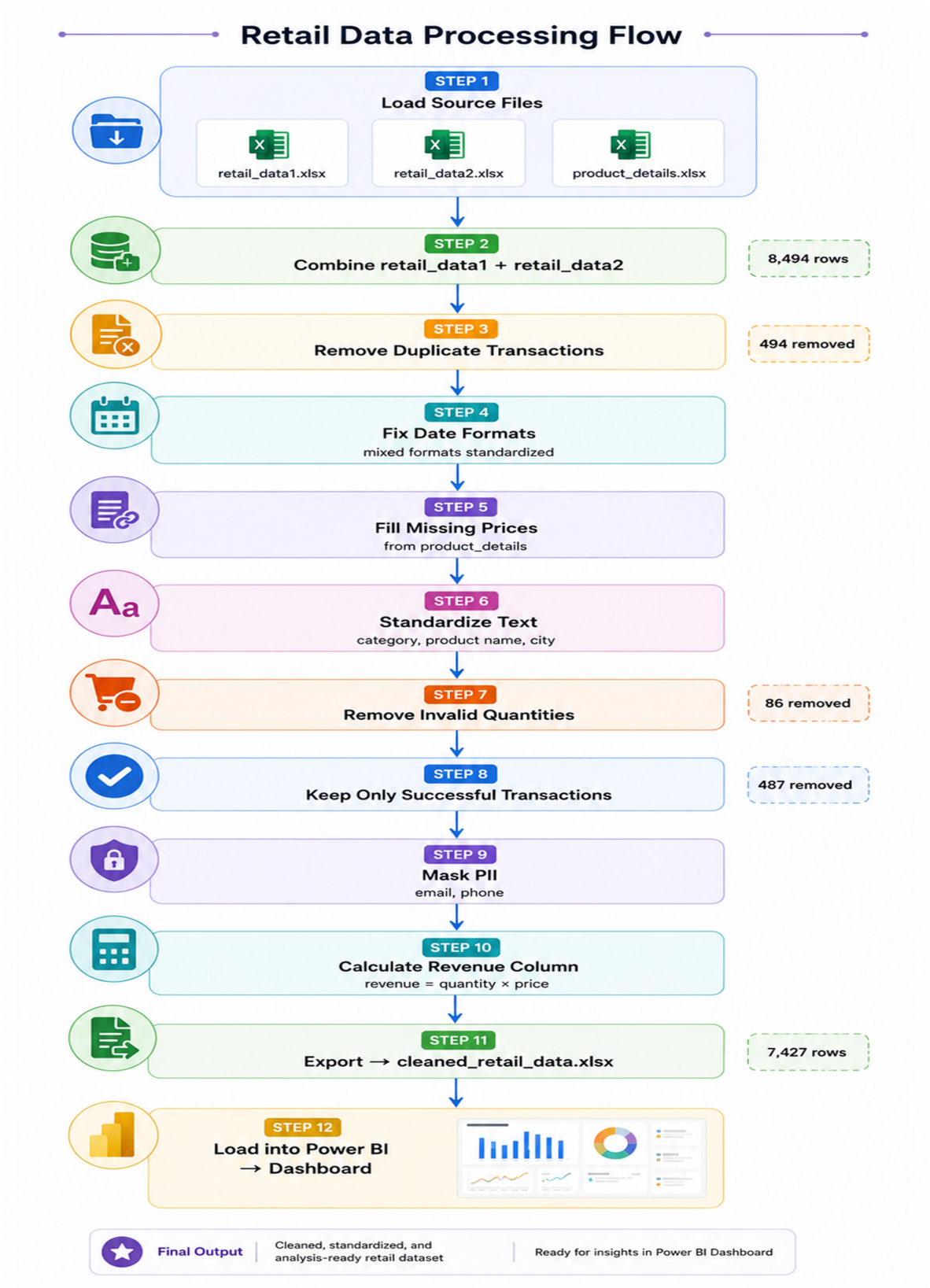
Section 1: Project Overview

ABC Retail Solutions operates across multiple cities through online and offline sales channels. The company collects transactional data from two source systems which contain inconsistencies such as duplicates, missing values, invalid quantities, and inconsistent naming conventions. This project builds an end-to-end data pipeline using Python that ingests, cleans, transforms, and loads retail transaction data into a cleaned dataset. The cleaned data is then visualized using a Power BI dashboard to generate business insights and KPIs.

Section 2: Architecture Diagram



Section 3: Data Flow Diagram



Section 4: Data Description

Dataset	Rows	Columns	Description
retail_data1	4,243	16	First transaction source system
retail_data2	4,251	16	Second transaction source system
product_details	10	4	Reference/dimension table
cleaned_retail_data	7,427	21	Final cleaned dataset

Section 5: Data Cleaning Strategy

Issue Found	Count	Fix Applied
Duplicate transactions	494	Removed, kept successful payment record
Missing prices	Several	Filled from product_details using product_id
Invalid quantities (0 or negative)	86	Rows dropped
Failed/pending transactions	487	Rows dropped
Mixed date formats	Multiple	Converted using pd.to_datetime
Inconsistent category names (ELEC, FURN, CLOTH)	Multiple	Mapped to standard names
Inconsistent product names (LAPTOP, laptop)	Multiple	Converted to Title Case
PII data (email, phone)	All rows	Email masked, phone hashed

Section 6: Transformation Logic

1. REVENUE CALCULATION

Formula: $\text{revenue} = \text{price} \times \text{quantity} \times (1 - \text{discount})$

Example: $\text{₹}70,000 \times 2 \times (1 - 0.10) = \text{₹}1,26,000$

2. PII MASKING

Email : First 2 characters shown, rest masked

Example: JohnDoe@gmail.com →
Jo****@gmail.com

Phone : Hashed using SHA-256 algorithm

Example: 9876543210 → a3f8c2d1e4

3. DATE STANDARDIZATION

Mixed formats like 7/21/2025 and 02-19-2026

All converted to standard format: YYYY-MM-DD

4. CATEGORY STANDARDIZATION

ELEC / electronics → Electronics

FURN / furniture → Furniture

CLOTH / clothing → Clothing

HOME / home appliances → Home Appliances

Section 7: KPI Summary

KPI	Value
Total Revenue	₹1,08,88,54,335
Total Orders	7,427
Average Order Value	₹1,46,607
Total Units Sold	22,125
Top Category	Electronics (₹62.72 Cr)
Top City	Chennai (₹23.28 Cr)
Top Product	Laptop (₹43.18 Cr)
Top Payment Method	Card

Section 8: Visual Insights

1. Electronics dominates revenue contributing 57% of total revenue, driven mainly by Laptop sales.
2. Chennai leads all cities in revenue followed closely by Delhi and Hyderabad.
3. Online and Offline channels are almost equal showing balanced sales across both channels.
4. Laptop is the single highest revenue product at ₹43.18 Crores, far ahead of all other products.
5. Card and Cash are the most preferred payment methods among customers.
6. Revenue peaks during certain months showing clear seasonal trends useful for planning promotions.

Section 9: Assumptions

1. Missing prices were filled using the product_details reference table based on product_id.
2. Where duplicate transaction_id existed with both successful and failed status, only the successful record was kept.
3. Rows with quantity = 0 or negative were considered invalid and removed.
4. Only transactions with payment_status = successful were included in revenue calculations.
5. Category names were standardized based on the product_details reference table.
6. All revenue figures are in Indian Rupees (₹).

Section 10: Conclusion

The end-to-end data pipeline successfully ingested, cleaned, and transformed 8,494 raw retail transactions into 7,427 high-quality records ready for analysis.

Key business insights reveal that Electronics is the top-performing category, Laptop is the bestselling product, and Chennai generates the highest regional revenue.

The Power BI dashboard provides interactive exploration of revenue trends, product performance, category analysis, and regional insights enabling data-driven decision making for ABC Retail Solutions.